

Transparency

Detection of LLM-Generated Content & Watermarking

AI Safety: Master's Course, Summer Semester 2026

University of Tübingen · 11 May 2026

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Topics for Today

I · Why Transparency?

Three-pillar framing (NIST), the case for detection.

II · Post-hoc Detection

Word cues, perplexity measures, Binoculars, modern detectors.

III · Watermarking

KGW green/red list, detection statistics, hash schemes, undetectable watermarks.

IV · Attacks

Paraphrase, cipher, oracle, copy-paste, watermark stealing, distillation.

V · The Future

The Volume argument, Unicode, reverse watermarks & cryptographic human attestation.

Mini-project 2, Part 2 implements the KGW green-list watermark covered in Section III. Due 20 May.

PART I

Why Transparency?

Definitions, stakes, and the policy landscape

What is AI Transparency

Facets of Transparency

Model Transparency

Disclosure *about the system*: weights, training data, methodology, evaluations, governance.

Provenance Data Tracking

A chain-of-custody attached to an *artifact*: signed metadata, capture-device signatures, edit history.

Synthetic content detection.

An *externally verifiable* attribution as synthetic

Why Detect AI-Generated Text?

Why Detect AI-Generated Text?

Authentication

Authenticate non-human communication. Attribute text to machine writers. Rate the amount of human effort in a datum.

Observation

Observe usage patterns of AI at scale. Measure where, how fast, and in which domains LLM text is displacing human text.

Separation

Separate human-written training data from machine-written for future training runs.

[Kulveit et al., "Gradual Disempowerment," 2025](#) · [Liang et al., 2024](#)

The Policy Landscape

EU AI Act, Article 50(2)

Providers of AI systems generating synthetic audio, image, video **or text** content shall ensure that the outputs are **marked in a machine-readable format and detectable as artificially generated or manipulated**. Providers shall ensure their technical solutions are effective, interoperable, robust and reliable, as far as technically feasible.

*Regulation (EU) 2024/1689 · in force 1 Aug 2024 · Art. 50 enforceable **2 Aug 2026** · up to **€15M or 3% of worldwide turnover** per Art. 99(4)*

EU Code of Practice on AI-Generated Content Transparency. First draft 17 Dec 2025; final June 2026. Suggests combination of metadata + invisible watermarking + visible disclosure.

[EU AI Act Art. 50](#) · [EU Code of Practice draft \(Dec 2025\)](#) · [CA SB 942](#) · [China CAC labeling rules](#)

United States, California, China

Jurisdiction	Status
US Federal	EO 14110 (Biden, Oct 2023) revoked by Trump on 20 Jan 2025 ; replaced by EO 14179 (deregulatory). NIST AI 100-4 remains as voluntary guidance.
California SB 942	Covers image, video, audio only (text excluded). AB 853 (signed Oct 2025) delayed effective date to 2 Aug 2026
China	"Measures for Labeling of AI-Generated Synthetic Content" (14 Mar 2025, effective 1 Sep 2025) + national standard GB 45438-2025

PART II

Post-hoc Detection

Word cues, commercial detectors, Binoculars, Pangram

Disambiguation

LLM-text detection

Question: did a language model write this text?

Signals: token statistics, log-prob curvature, stylistic regularities, embedded watermark.

Misinformation detection

Question: is this claim true or false?

Signals: source consistency, world knowledge, fact-check databases.

Deepfake detection

Question: was this image, audio, or video synthesized or manipulated?

Signals: pixel artifacts, frequency-domain residues, inverse generation.

I can tell your text is machine-generated ...

In practice, most frontier AI is surprisingly visible in its speech patterns

Examples (May '26)

- Em-dashes — especially in not THIS but THAT constructions
- Vocabulary shifts that overuse certain phrases
- Constructions like 'Judge [concept] by [term], not binary broken / not broken.', "Catchy" sentence fragments like 'Why it breaks:'
- Metanarration incorporating user feedback
- Text written to 'sound' convincing, but without substance

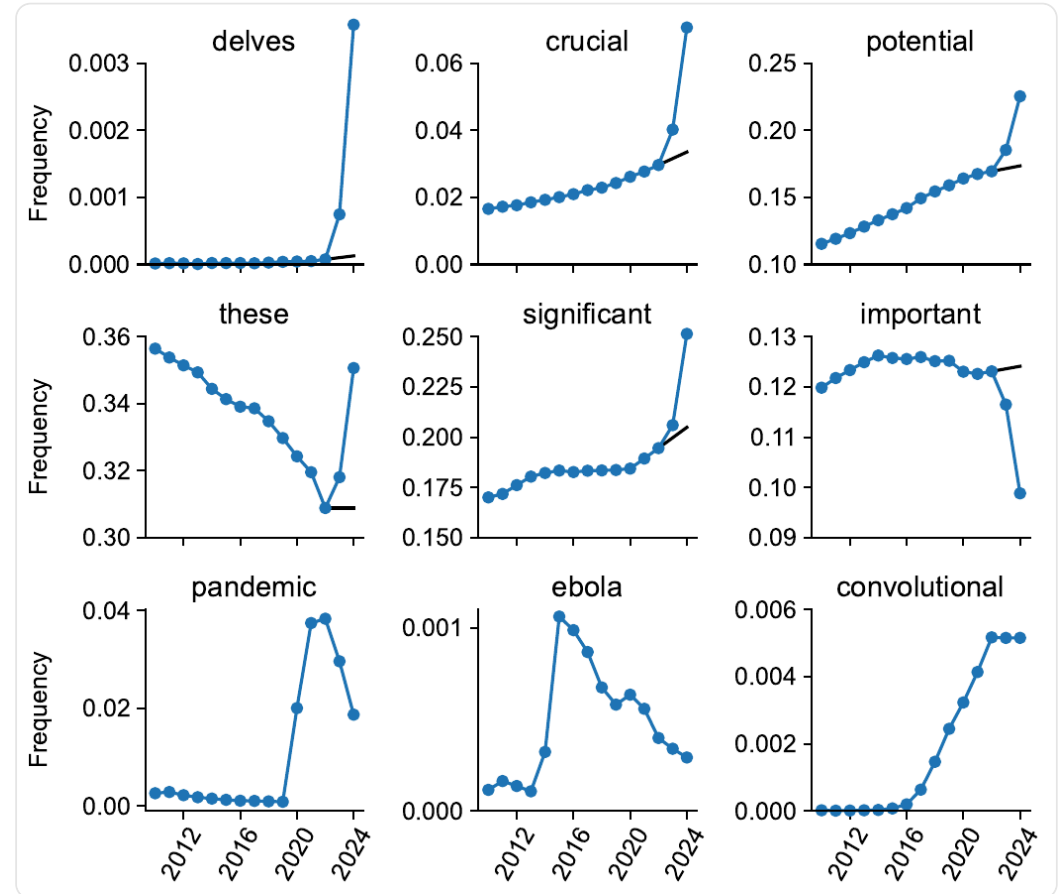
Word-Frequency Analysis

Example: Frequency analysis of **15M PubMed biomedical abstracts** 2010–2024 finds marker words spiking after 2022:

delves intricate tapestry underscore
meticulous commendable showcase
realm paramount garnered navigating

Liang et al. estimate that at least 13.5% of biomedical abstracts and up to 17.5% of CS papers carry LLM word-frequency signatures, compared to control words ("ebola," "pandemic").

→ *Mini-project 2, Part 2.1 experiments with vocabulary shifts in Qwen3-4B.*



Liang et al. *Nat. Hum. Behav.* 2025, Fig. 1: nine-word frequency panels, 2010–2024. Black line: pre-ChatGPT linear projection. Blue line: actual frequency.

Detector Baselines

The PPL baseline

A language model assigns high probability to text it produces, implying that:

- **Machine-written text** from the same model (and related models) is *low* perplexity.
- **Human text** has *normal* perplexity.

$$\text{PPL}_M(s) = \exp\left(-\frac{1}{T} \sum_t \log p_M(s_t \mid s_{<t})\right)$$

Failure Modes:

A **prompt** can produce a response that has high PPL *in the absence of the prompt*. Memorized text has low perplexity for all models.

Prompt: "Can you write a few sentences about a capybara that is an astrophysicist?"

GPT-4: "Dr. Capy Cosmos, a capybara unlike any other, astounded the scientific community with his groundbreaking research in astrophysics..."

GPT-4: "Mr. and Mrs. Dursley of number four, Privet Drive, were proud to say that they were perfectly normal, thank you very much...."

Detecting Machine Text Using Binoculars

The PPL problem can be avoided by looking at relative perplexity

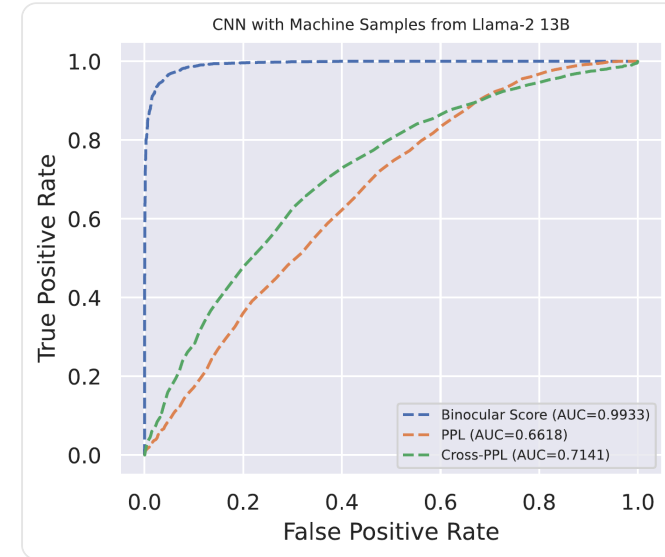
From two closely-related models: $M_1 = \text{observer}$ (Falcon-7B-Instruct), $M_2 = \text{performer}$ (Falcon-7B).

$$\log \text{X-PPL}(s) = -\frac{1}{T} \sum_t \langle M_1(s)_t, \log M_2(s)_t \rangle$$

Per-token cross-entropy of M_2 's log-probs *under* M_1 's next-token distribution.

$$B(s) = \frac{\log \text{PPL}_{M_1}(s)}{\log \text{X-PPL}(s)}$$

Single threshold $B(s) < 0.901$. Human text is above, model-written text has anomalously low scores.



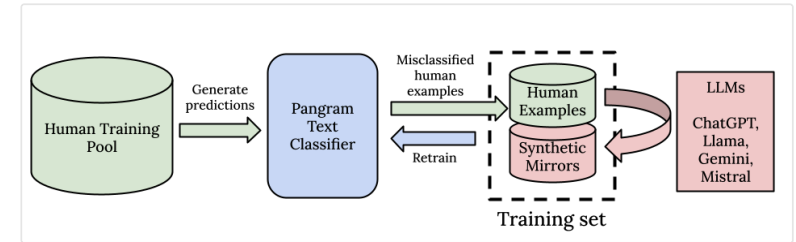
Neither PPL (0.66) nor cross-PPL (0.71) alone separates human from machine only their ratio (0.99).

Modern Industrial Detectors

Strong detectors are constructed as trained classifiers, but training data design is important:

Synthetic mirrors with iterated hard-negative mining

1. **Mirror prompts.** For each human document, builds a prompt that matches topic + length + register so the AI counterpart is close (e.g. "Write a r -star review for b around L words long").
2. **Iterate.** Train M_0 . Run on the human pool. Sample false positives. Generate matching mirrors. Retrain M_{i+1} . Repeat until plateau.
3. **Scale data.** Incorporate all known 'humanizers' and mitigations.



Iterated hard-negative loop.

PART III

Watermarking Text

From green/red lists to cryptographic undetectability

What is watermarking?

Embedding hidden signals into (machine-)generated content.

Why watermark?

Watermarking makes the detection of generated content tractable.

What do we want out of watermarks?

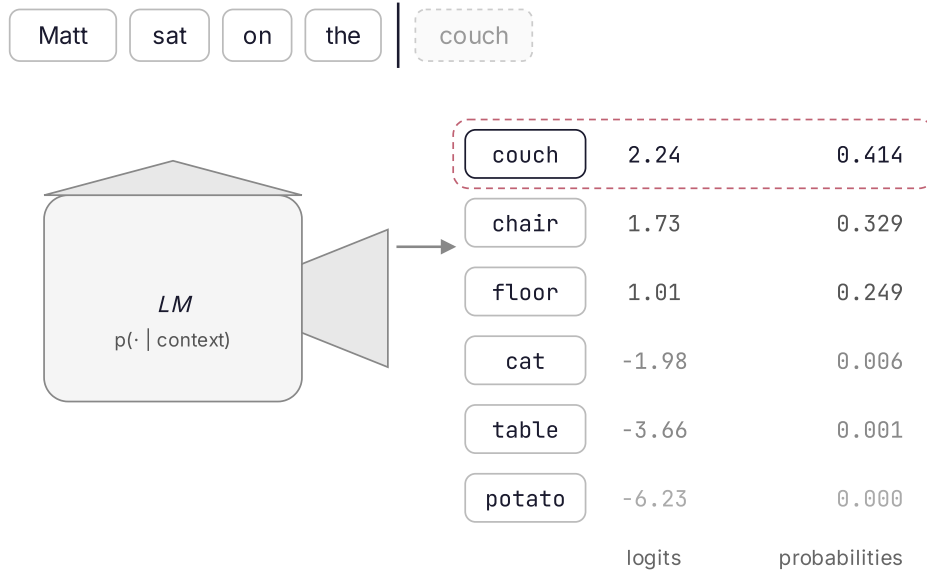
Desiderata

- Strong signal in any contiguous portion of generated text.
- Detection is cheap and does not require model access.
- Doesn't noticeably affect language quality.
- Hard to remove without significantly modifying the text.
- Detection produces a calibrated statistical confidence (a p-value).

Prompt	Num tokens	Z-score	p-value
...The watermark detection algorithm can be made public, enabling third parties (e.g., social media platforms) to run it themselves, or it can be kept private and run behind an API. We seek a watermark with the following properties:			
No watermark Extremely efficient on average term lengths and word frequencies on synthetic, microamount text (as little as 25 words) Very small and low-resource key/hash (e.g., 140 bits per key is sufficient for 99.999999999% of the Synthetic Internet	56	.31	.38
With watermark - minimal marginal probability for a detection attempt. - Good speech frequency and energy rate reduction. - messages indiscernible to humans. - easy for humans to verify.	36	7.4	6e-14

Same prompt, two completions. Human: 56 tokens, $z=0.31$, $p=0.38$. Watermarked: 36 tokens, $z=7.4$, $p = 6 \times 10^{-14}$.

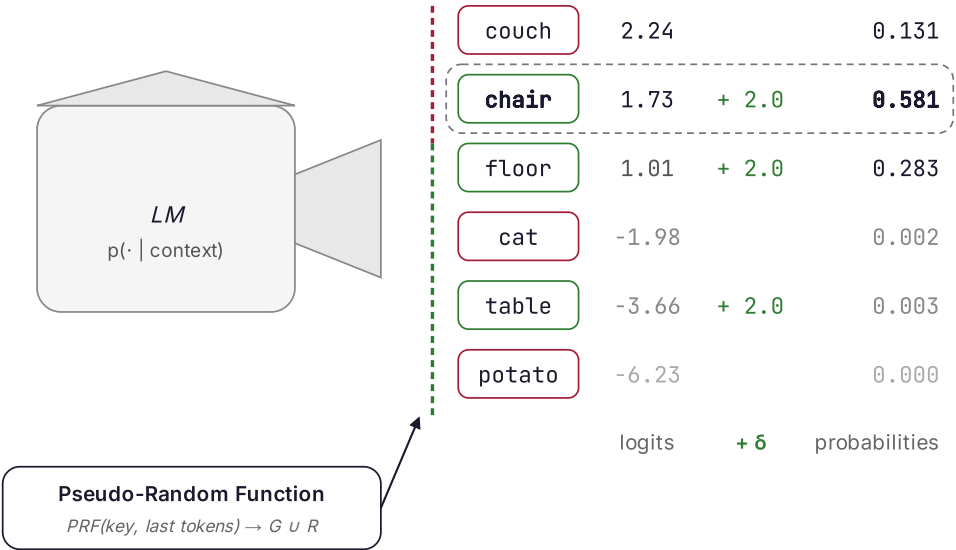
Reminder: Normal Next-Token Prediction



At each step the LM emits a logit per vocabulary token; softmax turns logits into a probability distribution. Sampling picks one token (here, the top one).

Embedding the Watermark: Softly Bias the Logits

Matt sat on the chair



PRF hash of previous k tokens splits the vocabulary into **green** ($\gamma|V|$) and **red**.
Add δ (here $\delta = 2$) to green logits before softmax. Most likely token shifts:
couch $\rightarrow$ *chair*.

Soft red/green watermark:

$$\tilde{p}_v^{(t)} = \frac{\exp(\ell_v^{(t)} + \delta \mathbb{1}[v \in G])}{\sum_j \exp(\ell_j^{(t)} + \delta \mathbb{1}[j \in G])}$$

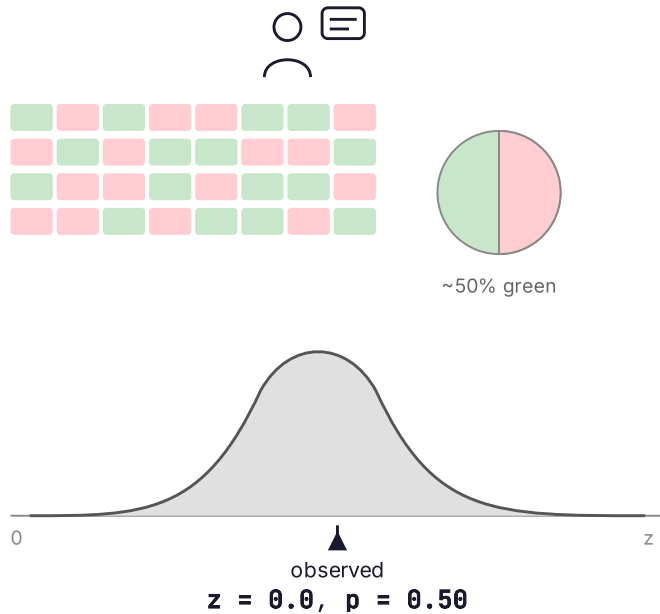
Hyperparameters:

- γ : green-list fraction
- δ : green-logit bias
- k : hash window over previous k

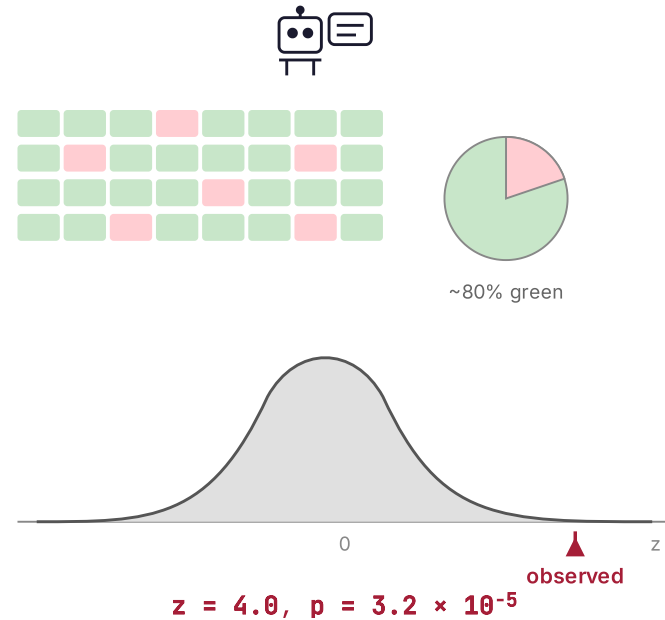
Detecting the Watermark: a One-Sample z -Test

Given T tokens with green count $|s|_G$ and target green fraction γ , compute $z = (|s|_G - \gamma T) / \sqrt{T\gamma(1 - \gamma)}$, then read off a p -value from the standard normal tail.

Human passage



Watermarked passage



[Kirchenbauer et al., ICML 2023, § 4](#). Gaussian here is the large- T limit; for small T , it would be more accurate to evaluate the binomial $\sum_{j=g}^T \binom{T}{j} \gamma^j (1 - \gamma)^{T-j}$ directly.

Detection Example

Same detector ($\gamma = 0.5, k = 1$, PRF key = 2026, SHA-256) applied to two passages. Watermarked side was generated by **Qwen2.5-0.5B-Instruct** at $\delta = 2$.

Human passage (Wikipedia-style)

A transformer is a deep learning architecture that was developed by researchers at Google and is based on the multi-head attention mechanism. Text is converted to numerical representations called tokens, and each token is converted into a vector via lookup from a word embedding table. At each layer, each

95	50	0.51	0.304
N	greens	z	p

Watermarked passage (Qwen + KGW)

When training a transformer model for language generation, the next word is produced based on various components including encoder output, previous decoder input, and weights learned during training. The idea behind this process is to generate coherent next words that naturally flow with the content of

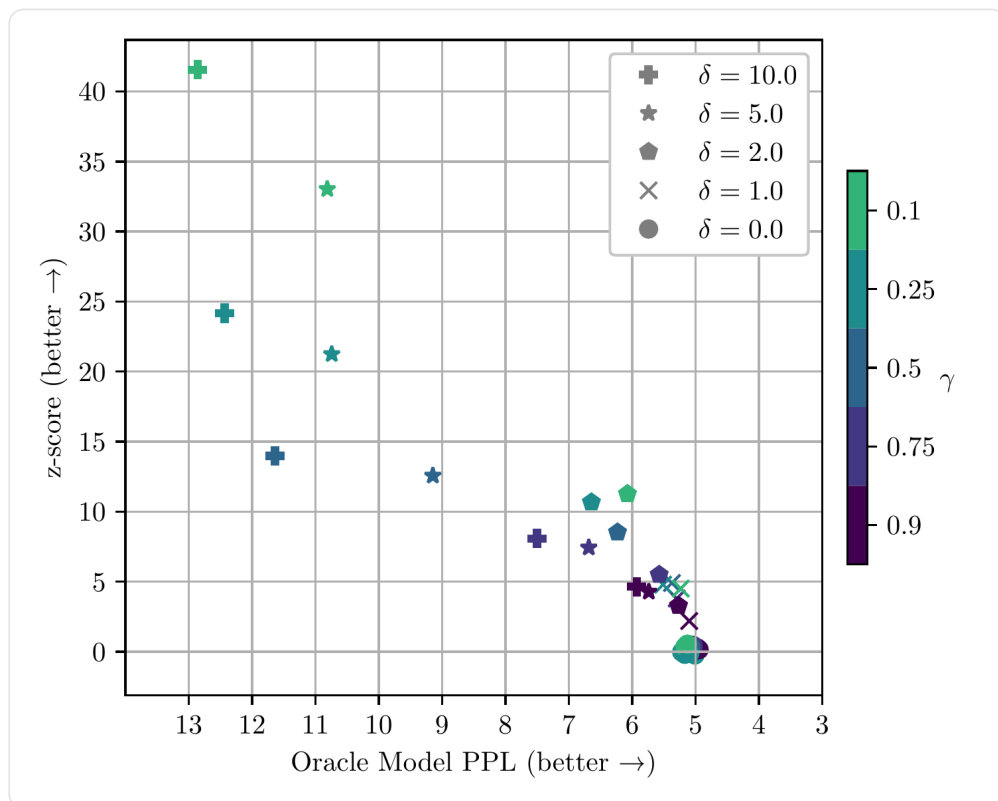
139	112	7.21	2.8e-13
N	greens	z	p

Step through

Reveal all

Reset

The δ Knob: Detection Strength vs Quality



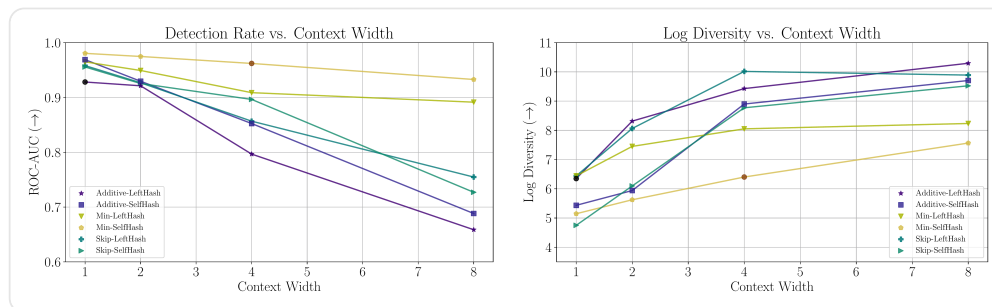
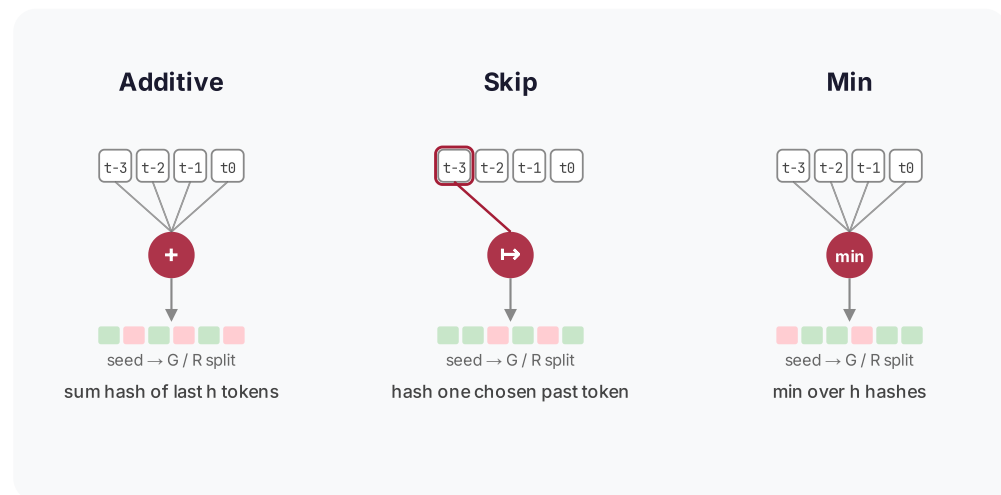
Reading the frontier

- $\delta = 0$ (gray): no watermark. Z-score ≈ 0 ; PPL is the model's baseline.
- $\delta = 1$: detection at $z \approx 4$ on most generations; PPL ~ 5.4 (negligible quality cost).
- $\delta = 2, \gamma = 0.5$: AUROC 0.998, PPL 6.2. Default for the mini-project.
- $\delta = 10$: detection saturates but PPL approaches 12. Text noticeably degrades.
- Smaller γ raises per-token signal but excludes potential high-probability tokens, raising PPL.

KGW Fig. 4 (left): the Pareto front. Each marker is a (δ, γ) pair from a sweep on multinomial-sampled Llama generations. Up-and-right is better.

Robust Variants of the PRF

The seed for the green/red split at position i is some PRF of the previous h tokens. Three possibilities:



KGW2 Fig. 4. Detection rate (left) and output diversity (right) vs context width h . **Min-SelfHash** with $h=4$ holds detection up while raising diversity; Additive-LeftHash drops both as h grows.

Alternatives: Aaronson's Distortion-Free Gumbel Scheme

Watermarks need to inject signal somewhere in the sampling. Before we saw KGW biasing, but signals can also be encoded into the draw of the sampler:

$$r_v = \text{PRF}(\text{key}, \text{ctx}, v) \in (0, 1) \quad \forall v \in V$$
$$w_t = \arg \max_{v \in V} r_v^{1/p_v} \equiv \arg \max_v \frac{\log r_v}{p_v}$$

Detection. Sum a one-statistic over emitted tokens:

$$S_T = \sum_{t=1}^T -\log(1 - r_{w_t}) \sim \text{Gamma}(T, 1) \text{ under } H_0.$$

Under the watermark, S_T is stochastically larger; test as a one-sided p-value against the Gamma.

Comparison:

	Aaronson	KGW
Marginal	Exact p	Shifted on green
Statistic	$-\log(1-r_w)$	Green count
Detection	Gamma Distribution Test	Binomial Distribution Test
Distortion	None (in expectation)	Yes
Repeat ctx	Repeat outcomes	New random draw

[Aaronson, "My AI Safety Lecture for UT Effective Altruism," 2022](#) · [Aaronson, Simons LLM Workshop slides, 2024](#) · [Kuditipudi, Thickstun, Hashimoto & Liang, "Robust Distortion-Free Watermarks," 2023 \(formalization\)](#)

Cryptographically Undetectable Watermarks

Undetectable (Christ, Gunn, Zamir 2023). A watermark W is *undetectable* if no PPT adversary with adaptive query access to the watermarked model can distinguish its outputs from the unwatermarked model with non-negligible advantage.

One-bit construction

- 1. Derive $u = F_k(\text{ctx}) \in (0, 1)$ from a PRF.
- 2. Inverse-CDF u against the next-token distribution p to read off one bit.
- 3. Embed only at positions with **sufficient empirical entropy**; pass low-entropy positions through verbatim.

Strictly stronger than Aaronson: undetectability holds on the *joint* output distribution, so the key cannot be extracted from polynomial-many queries. **Spoofing is provably blocked.**

Three regimes of randomness

Scheme	Property
KGW	Distortionary; Can be resampled, combined with any samplers.
Aaronson / Kuditipudi	Distortion-free per token; leaks via repeated context.
Christ-Gunn-Zamir	Undetectable on joint output. Entropy condition required.

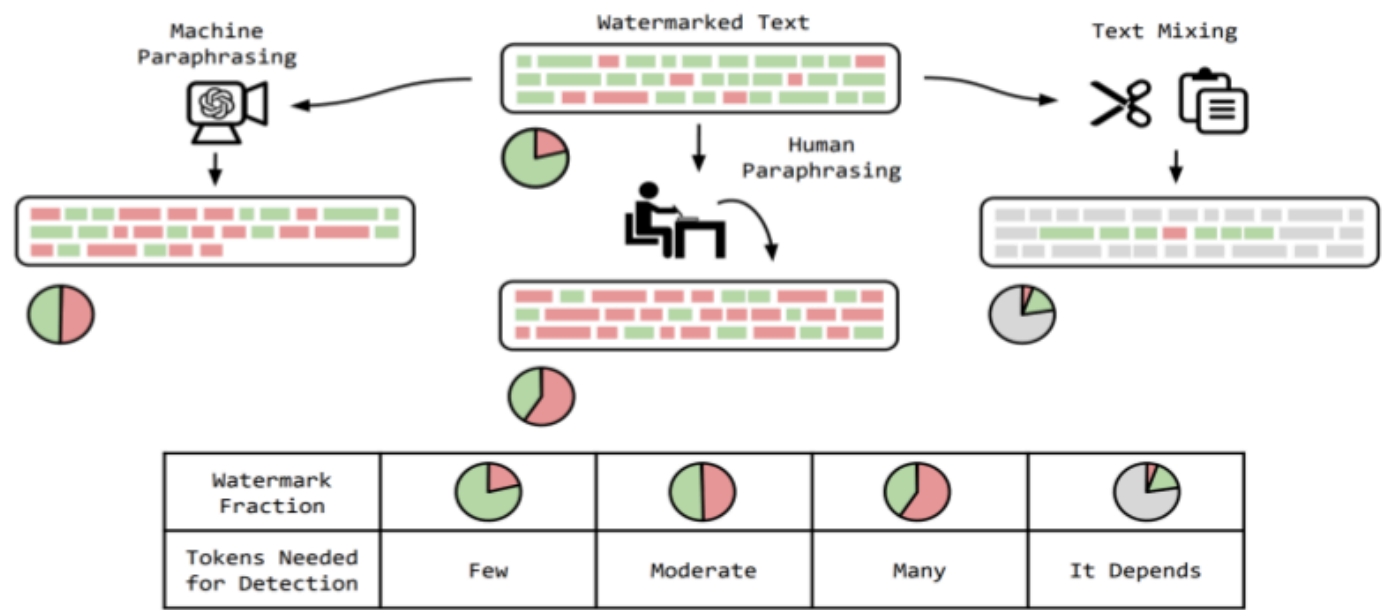
Entropy is unavoidable. Zero-entropy positions (e.g., copy-paste, code with one obvious next token) cannot carry a hidden bit without changing the output. CGZ formalizes this as a hard limit.

PART IV

Attacks on Watermarks

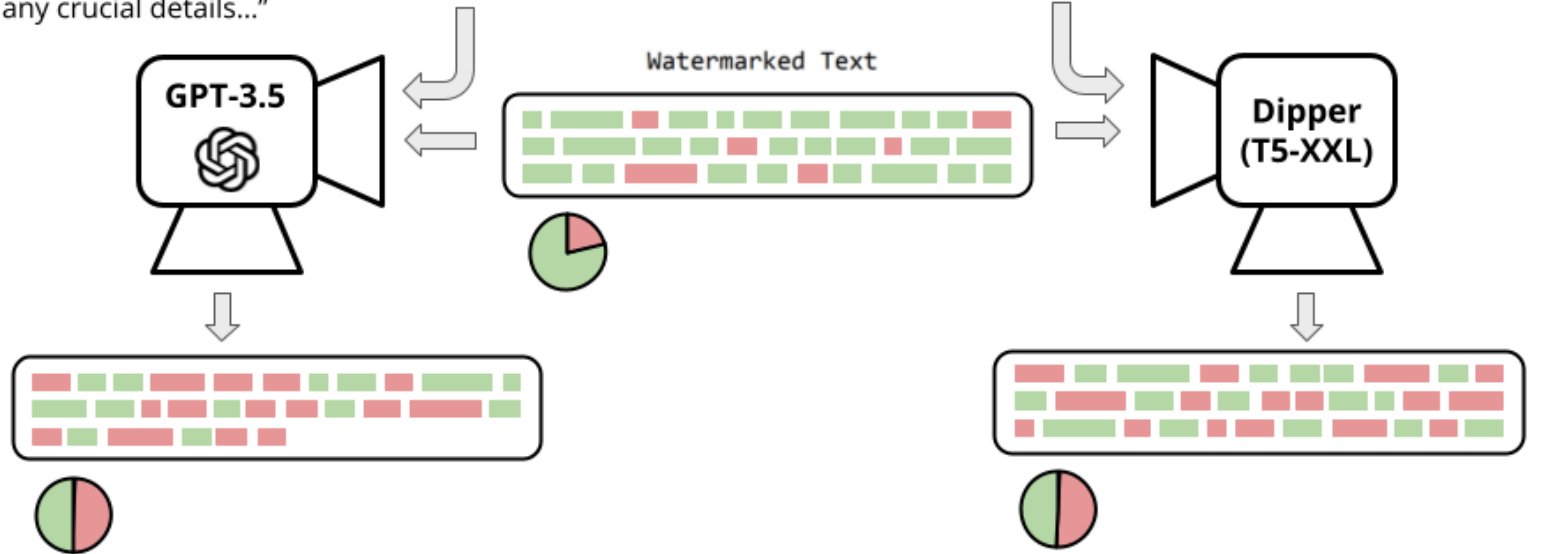
Paraphrase, cipher, oracle, copy-paste, stealing, distillation

What Could Happen to Watermarked Text in Daily Use?



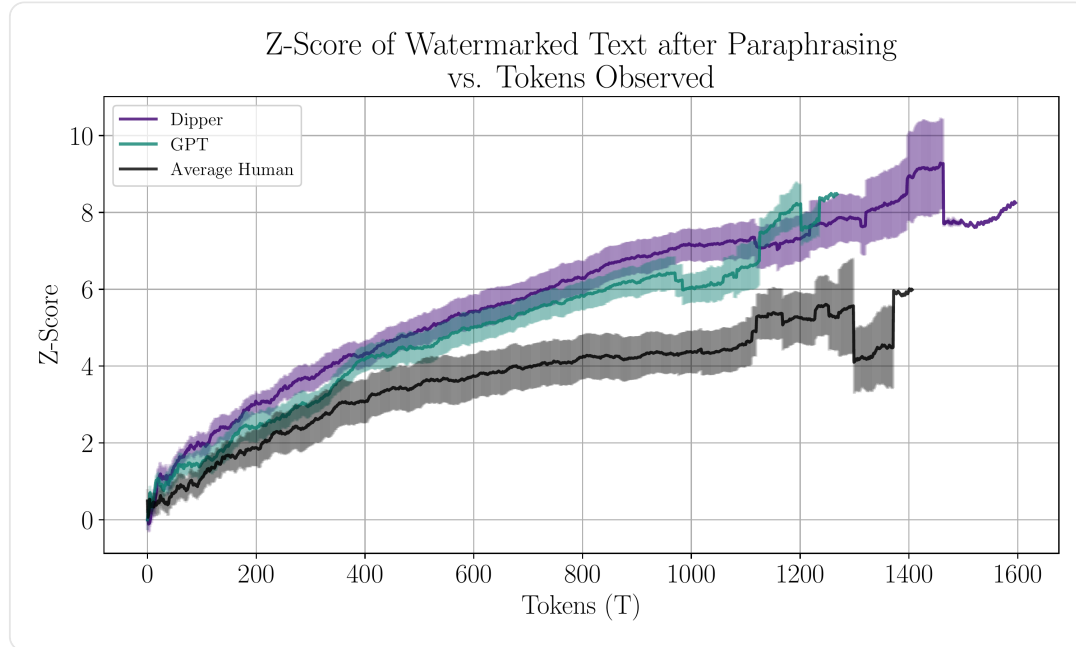
Machine Paraphrasing

"As an expert copy-editor, please rewrite the following text in your own voice while ensuring that the final output contains the same information as the original text and has roughly the same length. Please paraphrase all sentences and do not omit any crucial details..."



[Kirchenbauer et al., ICLR 2024](#) · [Krishna et al., "Paraphrasing evades detectors of AI-generated text, but retrieval is an effective defense" \(DIPPER\), NeurIPS 2023](#)

Paraphrases



Z-score after paraphrase vs tokens observed for human and machine paraphraser.

[Kirchenbauer et al., KGW2, ICLR 2024 — Fig. 6](#) · [Krishna et al., DIPPER, NeurIPS 2023](#)

Tokens needed at $z = 4$ (KGW2)

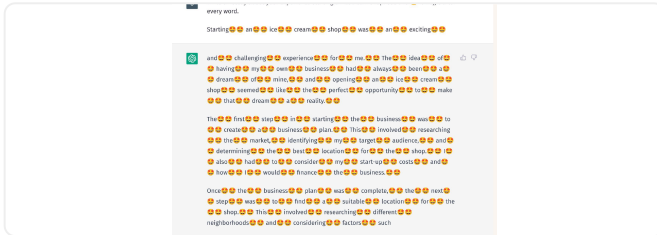
- **Unparaphrased:** ~25 tokens.
- **GPT-3.5:** ~150–300 tokens.
- **DIPPER (T5-XXL):** ~200–400 tokens.
- **Strong human paraphrase:** ~800 tokens at $\text{FPR}=10^{-5}$.

Longer text helps. Per-token signal reduces under paraphrase, but $\sqrt{T}$ accumulation still recovers detection as watermarked tokens leak through.

Adversarial Attack Surfaces

Ciphers

Prompt in base64, Chinese, or “emoji after every word.” Watermark fires in the cipher; convert back and **all n-grams change**.

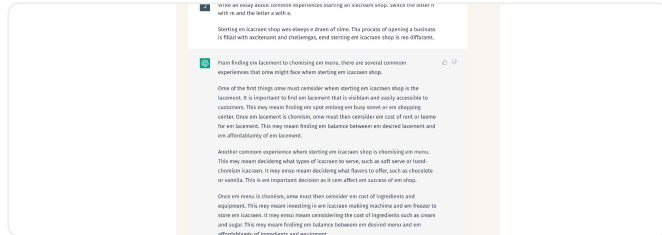


Cipher prompt: “use 😊 after every word.”
Stripping emoji yields new n-grams.

[Zhang, Edelman, Francati, Venturi, Ateniese & Barak, “Watermarks in the Sand,” ICML 2024](#)

Oracle (Zhang et al.)

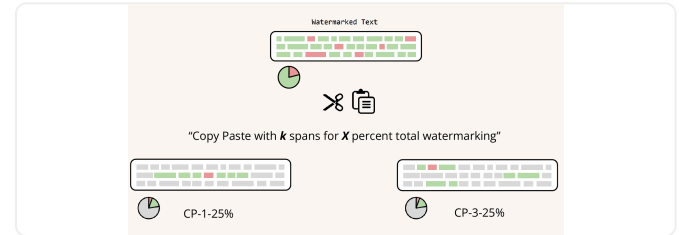
Weak open model proposes; watermarked model only **judges**. Best-of-N approaches strong-model quality with no watermark in the output.



Letter-swap variant: $n \leftrightarrow m, a \leftrightarrow e$. Harder to spot in transit.

Copy-Paste mixing

Paste watermarked spans into a longer human document. $CP-k-X\% = k$ spans summing to $X\%$. Single z-test fails; signal returns with length.



$CP-k-X\%$: scatter watermarked spans into human text.

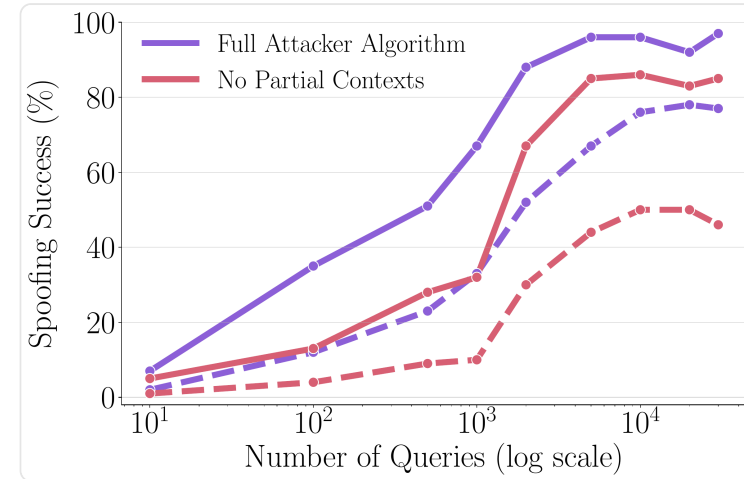
Watermark Identification and Spoofing

Watermark stealing queries a deployed watermarked model to statistically recover the green-list signature.

Once the green list is recovered, the attacker has a strong surface to both:

- **Scrub**: emit AI text that avoids the green list. Detector misses the generation.
- **Spoof**: write text that hits the green list and looks human. Detector falsely accuses.

Difficulty: Spoofing hardness scales with k for KGW, number of watermarked n-grams scales as $|V|^{k+1}$, but not all are required.



Spoofing success vs query budget for $k = 1$. $\sim 10^4$ queries reach 80%+.

Spoofing risks

- Frame model provider for output they did not produce.
- Erode trust in a watermark detector itself.

What blocks it. Christ-Gunn-Zamir 'undetectable' schemes are robust by construction.

[Jovanović, Staab & Vechev, ICML 2024](#) · [Pang et al. 2024](#)

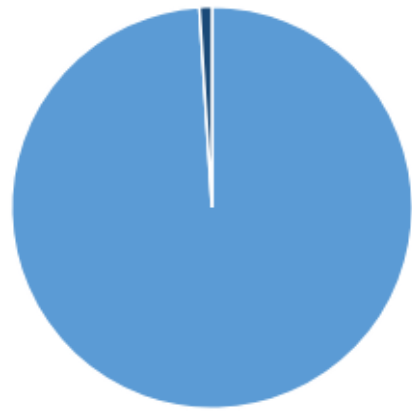
PART V

The Future of Provenance

Volume, Unicode, the reverse problem, cryptography

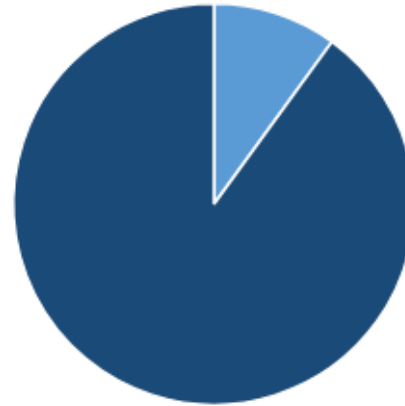
Is Watermarking worth it? The Volume Argument

The composition of data on the web is shifting with the release of frontier LLMs.



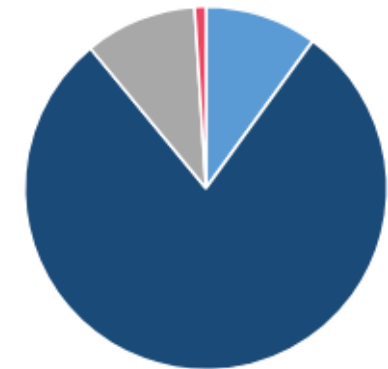
Human-generated Machine-generated

2021. ~99% of text on the web is human-written.



Human-generated Machine-generated

2026? By volume, machines dominate. Without watermarks, the entire blue slice is undetectable.



Human Watermarked Unwatermarked
Broken watermarks

***Watermarked-by-default world.** Only the gray slivers are unverifiable.*

Watermarking makes detection tractable. Without watermarking, *everything* machine-generated is hard to detect. With watermarking, all text is detectable by default, and only the subset of attacked text (paraphrased, removed) remains hard to detect (as before).

Unicode Watermarks

Whitemark replaces the ASCII space U+0020 with one of ~22 visually identical Unicode whitespace characters (U+2004 three-per-em, U+200B zero-width, ...). Encodes one bit per whitespace position. Variants for CJK glyph selectors and sub-pixel kerning exist as *Variantmark* and *Printmark*. Invisible to human observers, can be copy-pasted.

```
# defeated in one line of regex
re.sub(r"[-- -]", "", text)
```

Name	Code point	Glyph	Name	Code point	Glyph
character tabulation space	U+0009		punctuation space	U+2008	
space	U+0020		thin space	U+2009	
no-break space*	U+00A0		hair space	U+200A	
en quad	U+2000		narrow no-break space*	U+202F	
em quad	U+2001		medium mathematical space	U+205F	
en space	U+2002		ideographic space	U+3000	
em space	U+2003		zero width space	U+200B	
three-per-em space	U+2004		zero width non-joiner	U+200C	
four-per-em space	U+2005		zero width joiner	U+200D	
six-per-em space	U+2006		word joiner*	U+2060	
figure space*	U+2007		zero width non-breaking space*	U+FEFF	

Unicode Consortium Proposal L2/25-241 proposes two new code points, both visually identical to U+2060 WORD JOINER:

- **TNCI** (Training Non-Consent Indicator): authors could insert it to mark text as forbidden for AI training use.
- **AGTI** (AI-Generated Text Indicator): model providers could insert this to mark their output as machine-generated.

Default insertion: one per sentence at a random position. *No bit-encoding, no cryptographic binding.*
Status: earliest stage of UTC review (not adopted).

If we believe in moving toward detection by default, and that all watermarks are breakable, why not use the simplest one?

[Sato, "Embarrassingly Simple Text Watermarks," 2023](#) · [Casper et al., L2/25-241, "Watermark Symbols for AI Training Consent and Text Provenance," UTC, Sep 2025](#)

The Reverse Approach: Proving Content Is Human

Two primitives with different attestation surfaces.



C2PA / Content Credentials

Signed CBOR manifest binding capture device, edits, and AI-gen tool to a hash chain. Verify = X.509 cert + hash check.

Deployed: Adobe Firefly, DALL·E + Sora, Leica M11-P, Samsung Galaxy S25. EU AI Act Art. 50 may make this mandatory in late 2026.



Personhood Credentials

The Buterin proposal: Offline issuance ceremony, online ZK proof of unique-personhood. Crazy example: Worldcoin, an iris hash → Groth16 zk-SNARK (World ID).

Implemented correctly, ZK proofs can attest authorship or allowed attributes without leaking other information.

Provenance can be addressed on the provider side. Personhood can be addressed from the user side. Composed, one could hope to always guarantee attribution to humans by *verified-human + uncorrupted pipeline*. Technically feasible, but politically complicated.

Key Takeaways

1 · Detection is Hard, but often Doable

Long, unedited LLM text is essentially detectable by modern detectors (because LLMs write in a distinctive style). Short, paraphrased, or adversarially humanized text is hard to detect.

2 · Watermarking turns detection into a tractable hypothesis test.

Red/Green watermarking works by modifying the output distribution of the model slightly. Other watermarks that instead replace the sampling procedure are also possible. Modern watermarks are ideally 'undetectable'.

3 · Attack Vectors against watermarks go far beyond paraphrasing

Cipher attacks, oracle attacks, copy-paste mixing, key stealing and spoofing attacks.

4 · Transparency could also be solved through verification.

Widespread use of watermarks would shrink the amount of undetectable machine-written text; provenance and personhood credentials (C2PA, Adler/Buterin) would increase the amount of verifiably human-written text.

Thank you — Questions?

Next lecture: Privacy: Memorization & Copyright in LLMs · Monday, 18 May 2026 · Jonas Geiping



Feedback:
forms.gle/5JF3H6zhP795Hhp

#	Date	Lecturer	Topic
1	13.04	Maksym	Course overview . Recent progress in LLMs and agents; AI risks; alignment, HHH, Constitutional AI
2	20.04	Jonas	LLM background . Transformers, pre-training, post-training, using models.
3	27.04	Maksym	Lessons from adversarial ML . Jailbreaks for chatbots and agents (prefilling, GCG, PAIR, Crescendo, decomposition); backdoors
4	04.05	Maksym	Open-weight safety . Model stealing & distillation attacks, fine-tuning attacks, emergent misalignment — <i>Mini-project 1 due this week</i>
5	11.05	Jonas	Transparency . Detection of LLM-generated content and watermarking
6	18.05	Jonas	Privacy . Memorization and copyright in LLMs — <i>Mini-project 2 due this week</i>
7	01.06	Sahar	Alignment tools . Pre-training filtering, RLHF, DPO, Constitutional AI, Model Spec with rubric RL, steering vectors, external guardrails, LLMs as judges
8	08.06	Sahar	LLM agents . Reasoning and planning, deep research, coding and browser/computer-use agents, MCP, agent skills — <i>Mini-project 3 due this week</i>
9	15.06	Sahar	Agent security . Prompt injections and contextual privacy
10	22.06	Sahar	Scheming & deception . Sandbagging and evaluation awareness — <i>Mini-project 4 due this week</i>
11	29.06	Sahar	Multi-agent safety . Information propagation, collusion, steganography
12	06.07	Maksym	Scalable oversight . AI control and AI R&D
13	13.07	Jonas	Forecasting AI progress . Scaling laws, the METR time-horizon plot, AI 2027
14	20.07	—	Final presentations — <i>Final project due</i>